

Banula Kumarage

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Summary

Innovative and skillful software engineer and researcher with experience in distributed systems, cloud-native technologies, and scalable backend development. Specializing in Java(Spring Boot), Kubernetes, and real-time data processing technologies such as Apache Beam and Kafka. Skilled professional bridging academic research and industry expertise to deliver effective solutions.

Working Experience

Software Engineer, Sysco LABS Sri Lanka

June 2024 – Present

- Designed and developed a real-time ETL pipeline using Apache Beam to streamline product catalog building by consuming events from multiple Kafka topics, transforming records, persisting them in a database, and publishing the processed data to relevant destination topics.
- Led the Kinesis-to-Kafka migration initiative by planning and implementing necessary changes, and successfully executing the production cutover without downtime.
- Executed over 20 production and staging deployments with zero rollback incidents, also responded and resolved critical production incidents, consistently meeting SLA requirements.
- Improved system performance and security by implementing AWS VPC Endpoints for private connectivity to AWS services, and enabled automatic scaling of Apache Flink applications across environments to optimize resource usage during peak demand.
- Enhanced application reliability and code quality by integrating TestContainers into CI/CD pipelines for isolated integration testing, while also addressing Kenna vulnerabilities, Veracode findings, and Sonar violations.
- Conducted in-depth analysis to identify cost-saving opportunities and performance bottlenecks, resulting in measurable efficiency improvements and reduced operational expenses.

Software Engineer Intern, IFS R&D International (Pvt.) Ltd

Jan 2023 – June 2023

- Automated the manual translation rejection sync process by developing a Jenkins pipeline, reducing execution time from nearly a day to just 15 minutes.
- Investigated custom autoscaling for IFS services on Kubernetes by identifying relevant dynamic metrics using Prometheus, Grafana, Postman, and K6 load testing.
- Optimized database performance 30% through implementing a priority-based caching algorithm in the PL/SQL layer to reduce overhead.
- Implemented graceful shutdown in Java microservices for Kubernetes pods using lifecycle hooks and Dockerfile modifications, improving autoscaling efficiency and preventing request loss.
- Actively contributed to fixing bugs in production services and enhanced code quality by writing unit and integration tests to increase overall test coverage.

Education

Master of Science (Major Component of Research), Reading

2025 - Present

Department of Computer Science and Engineering, University of Moratuwa

- Focusing the research on mitigating the cold start problem in Serverless Platforms
- Implementing and benchmarking the proposed solution using Apache OpenWhisk.

BSc. Eng. (Hons) Specialized in Computer Science & Engineering

2020 - 2024

Department of Computer Science and Engineering, University of Moratuwa

- Overall GPA: First Class GPA of 3.96 out of 4.2
- Dean's List in semesters 1, 2, 3, 4, 5, 6, 7, 8

Central College, Anuradhapura

- Island Rank - 173rd
- District Rank - 4th
- Z-Score - 2.2588

Publications

Use of Federated Learning for Personal Smart Media Cloud Solutions

Aug 2004

Banula Kumara, Hiruna Harankahadeniya, Akila Induranga, Indika Perera, Kutila Gunasekera

10.1109/MERCon63886.2024.10688883

Projects

PicsSmart : Privacy Ensured Smart Media Cloud with Federated Learning

July 2023 - June 2024

- An edge-processing-based application where users are concerned about their photos' privacy can still benefit from ML-powered smart features such as face-identification, face-clustering, and image caption generation. For the models to work well with heterogeneous datasets, models are to be frequently trained at the edge and updated using Federated Transfer Learning techniques.
- **Contribution:** Developing backend server with smart features, Federated Learning
- **Technologies:** FastAPI, ReactJS, LAVIS, Keras, Tensorflow Federated, Docker

Scatterrr : Secure and trusted distributed file storage system

April 2024 - July 2024

- A secure distributed file storage system that segments files into equal-sized chunks and distributes them randomly across a cluster of nodes for enhanced security. Using a Merkle Tree, it ensures data integrity, allowing users to verify files remain tamper-free. The system offers an intuitive interface for uploading, searching, viewing metadata, verifying integrity, and downloading files, combining reliability with robust security principles.
- **Contribution:** Back-end Developing
- **Technologies:** Java(Spring Boot), TypeScript(Next.js)

Streamix : Dash compliant video streaming system

Nov 2022 - Dec 2022

- The project is a dash (Dynamic Adaptive Streaming over HTTP) video streaming system. Android app is used to capture the video and upload it to the centralized server. Anyone connected to the server can watch the uploaded videos with different qualities.
- **Contribution:** Back-end Developing
- **Technologies:** Java, Node.JS, MongoDB, FFmpeg

Moodle plugin for establishing workflows

Sep 2022 - Nov 2022

- The purpose of the project is to create a plugin to automate the process of submitting and validating student requests regarding various Moodle activities.
- **Contribution:** As both a Back-end and Front-end developer
- **Technologies:** PHP, MySQL, JS, Mustache

Supply Chain Management System

June 2022 - July 2022

- The project is an information and data management system based on the distribution flow of a company. The complete chain of the distribution flow from the store to the customer can be monitored using the system.
- **Contribution:** As both a Back-end and Front-end developer
- **Technologies:** PHP, MySQL, JS, HTML

Pharmacy Management System - Home Pharmacist

Dec 2021 - Jan 2022

- A centralized system for pharmacies to provide many facilities including searching and ordering from pharmacies that are available in the system.
- **Contribution:** As both a Back-end and Front-end developer
- **Technologies:** HTML, CSS, PHP, MySQL

Student Request Management System

Dec 2021 - Jan 2022

- A web-based management system to manage student requests which have been sent to the academic staff in the university.
- **Contribution:** Back-end Developing
- **Technologies:** NodeJ.S, Express, React, MongoDB

Skills

Technical Skills

- Java | Python | Scala | Javascript | Typescript | PHP
- Spring Boot | Django | Quarkus | FastAPI | Node.js | Express | React JS
- MySQL | MongoDB | PostgreSQL | PL/SQL | SQLite
- Kubernetes | Helm | Docker | Kafka | Apache Beam | Redis | Elasticsearch
- AWS | Azure | GCP

Non Technical Skills

- Leadership | Teamwork | Communication | Public Speaking | Analytical Thinking

Certifications

Introduction to GCP Infrastructure 2024

- Obtained a verified badge after completing the lab tasks offered by Google Cloud

Application Development using Microservices and Serverless 2024

- Obtained a verified badge after completing the course offered by Coursera

AZ-900: Microsoft Azure Fundamentals Certification 2023

- Obtained a verified certificate after completing the certification exam offered by Microsoft

Kubernetes Certified Application Developer (CKAD) with Tests 2023

- Completed a course on the Udemy platform offered by KodeCloud

Neural Networks and Deep Learning 2023

- Completed a course on the Coursera platform offered by DeepLearning.AI

Introduction to Data Science in Python 2021

- Completed a course on the Coursera platform offered by the University of Michigan.

Achievements

Virtual Colleagues' Choice Award for the FY25 Sysco Technology Hackathon Oct 2024

- Our team ByteChef has won the Virtual Colleagues Choice Award in the Sysco Global Hackathon 2024! Also, we've been selected as a Top 5 finalist for our Generative AI-based solution for boosting the productivity of our sales reps.

Winners - RealHack 5.0 April 2024

- In the 24-hour hackathon organized by the University of Kelaniya, our implemented system of Multi-modal authenticity validation system for virtual interviews secured first place.

IEEEExtreme 16.0 - 24-hour Algorithmic Coding Competition Oct 2022

- Team404 | World Rank – 300 | Country Rank – 12

Responsibilities

Teaching Assistant

- Computer Networks (CS3033), Data Communication (CS2033) - Department of Computer Science & Engineering, University of Moratuwa

Deputy Head - Industrial Relations Committee and Department Coordinator

- EXMO 2023 Technical Exhibition organized by the University of Moratuwa

Field Representative

- Department of Computer Science and Engineering - batch 19 - Semester 3&4

Organizations

Toastmasters International

03/2019 - Present

- Member no: 7317863

Institute of Electrical and Electronic Engineers (IEEE)

2021 - Present

References

- *Available on Request*